

Project Initialization and Planning Phase

Date	28 June 2026
Team ID	xxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Define Problem Statements (Customer Problem Statement Template):

This template defines the major customer problems addressed by the OptiCrop system. The AI/ML solution analyzes soil nutrients (N, P, K), pH, temperature, humidity, and rainfall to provide accurate crop recommendations and support sustainable agriculture.

Problem Statement	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Farmer	Choose the best crop for my land.	I don't know which crop suits my soil and climate.	Soil nutrients, pH, rainfall, humidity and temperature keep changing.	To maximize the profit
PS-2	Agricultural Researcher	Analyze crop–environment relationships.	Large agricultural datasets are difficult to analyze manually.	Hidden patterns require AI and Machine Learning.	Confident in making accurate research decisions.
PS-3	Policy Maker	Plan sustainable agricultural policies.	Resource allocation and crop planning are complex.	Reliable predictive insights are limited.	Confident in planning evidence-based agricultural strategies.